HR Internal Operations Platform

REST API Reference Document

Version 1.0 · Acme People Ops · MERN Stack

A = Admin	H = HR	HM = Hiring Manager	I = Interviewer	C = Candidate	In = Intern	E = Employee

0. Auth & Session

Base URL: /api/v1 · All protected routes require Authorization: Bearer <JWT>

Method	Endpoint	Access	Description
POST	/auth/login	Public	Authenticate with email + password; returns signed JWT and role
POST	/auth/logout	All	Invalidate current JWT / clear session cookie
GET	/auth/me	All	Return current logged-in user profile and role
PATCH	/auth/me/password	All	Change own password (requires current password for verification)
POST	/auth/forgot-password	Public	Send password-reset email with expiring one-time token
POST	/auth/reset-password	Public	Reset password using the token received in email

1. Admin APIs

Admin has full system access. Role: A (Admin)

1.1 Users & Role Management

Method	Endpoint	Access	Description
GET	/users	A	List all user accounts with role, status, and active flag
POST	/users	A	Create a new user with bcrypt-hashed password and assigned role
GET	/users/:id	A	Get a single user's full profile and current role
PATCH	/users/:id	A	Update user name, email, or role assignment
PATCH	/users/:id/activate	A	Re-activate a previously deactivated user account
PATCH	/users/:id/deactivate	A	Soft-deactivate user account (blocks login, retains all data)
DELETE	/users/:id	A	Hard-delete user (only orphaned/test accounts; audit-logged)

1.2 Approval Workflow Configuration

Method	Endpoint	Access	Description
GET	/workflows	A	List all configured approval workflow chains
POST	/workflows	A	Create a new approval chain for a document type (offer/letter/exit)
GET	/workflows/:id	A	Get full step-by-step config for a specific workflow
GET	/workflows/type/:type	A, H	Fetch active workflow config by type (offer, letter, exit)
PATCH	/workflows/:id	A	Update step order, add/remove steps, or change approver roles
DELETE	/workflows/:id	A	Delete workflow config (only if no pending approvals exist on it)

1.3 Letter Template Management

Method	Endpoint	Access	Description
GET	/letter-templates	A, H	List all active letter templates with type and

			placeholder info
POST	`/letter-templates`	A	Create a new HTML template with {{placeholder}} tokens
GET	`/letter-templates/:id`	A, H	Get full template HTML content and placeholder list
PATCH	`/letter-templates/:id`	A	Edit template content, name, or type (auto-increments version)
PATCH	`/letter-templates/:id/deactivate`	A	Soft-deactivate template (hides from HR generator, retains history)
DELETE	`/letter-templates/:id`	A	Delete template (only if no letters have been generated from it)

1.4 Audit Log

Method	Endpoint	Access	Description
GET	`/audit`	A, H	List audit log entries with filters (module, action, actor, date range)
GET	`/audit/:id`	A, H	Get full details of a single audit entry including old and new values
GET	`/audit/export`	A, H	Export audit log as CSV or Excel (?format=csv|excel)

1.5 Admin Dashboard

Method	Endpoint	Access	Description
GET	`/dashboard/admin`	A	Return headcount, open candidates, offers in flight, active users, system health
GET	`/dashboard/admin/pipeline`	A	Return hiring pipeline stage distribution (Applied/Screening/Interview/Selected/Rejected)

1.6 Global Search & Export

Method	Endpoint	Access	Description
GET	`/search`	A, H	Full-scope search across Candidates, Employees, Offers (role-scoped results)
GET	`/search/export`	A, H	Export search results as CSV or Excel file

2. HR APIs

HR manages the full people lifecycle. Salary and PII fields are visible to HR. Role: H

2.1 Candidate Management

Method	Endpoint	Access	Description
GET	/candidates	A, H	List all candidates with pipeline status, position, and source
POST	/candidates	H	Create a new candidate record and generate their login account
GET	/candidates/:id	A, H, HM, I	Get candidate profile (salary/offer fields stripped for HM and I)
PATCH	/candidates/:id	H	Update candidate personal info, position, or application source
PATCH	/candidates/:id/status	H	Move candidate to next pipeline stage (Applied → Screening → Interview → Selected → Rejected)
PATCH	/candidates/:id/select	H	Mark candidate as Selected and set selectedAs (intern or employee)
PATCH	/candidates/:id/archive	H, A	Archive candidate (soft-delete; hides from active pipeline)
DELETE	/candidates/:id	A	Hard-delete candidate record (admin only; audit-logged)

2.2 Interview Scheduling

Method	Endpoint	Access	Description
GET	/candidates/:id/interviews	A, H, HM, I	List all interview rounds for a candidate
POST	/candidates/:id/interviews	H	Schedule a new interview round, assign Interviewer, and set date/time
GET	/candidates/:id/interviews/:roundId	A, H, HM, I	Get details of a specific round including submitted feedback
PATCH	/candidates/:id/interviews/:roundId	H	Update round date, time, or assigned Interviewer
DELETE	/candidates/:id/interviews/:roundId	H	Cancel/delete an interview round

2.3 Intern & Employee Directory

Method	Endpoint	Access	Description
GET	/employees	A, H	List all interns and employees with department, designation, and type
GET	/employees/:id	A, H, In*, E*	Get full profile (salary/PII visible to HR, Admin, and self only)
PATCH	/employees/:id	H	Update employee department, designation, manager, or status
POST	/employees/:id/history	H	Record a promotion, transfer, or increment with old/new values and reason
GET	/employees/:id/history	A, H, E*, In*	List full tenure history (promotions, transfers, increments)
PATCH	/employees/:id/status	H	Update employment status (active → notice → exited)

2.4 Offer Management

Method	Endpoint	Access	Description
GET	/offers	A, H	List all offers with status filter tabs
POST	/offers	H	Create a new offer auto-filled from candidate profile (status = Draft)
GET	/offers/:id	A, H, HM, C	Get offer details including CTC, designation, joining date, and template
PATCH	/offers/:id	H	Edit offer draft fields (salary, designation, joining date, template)
PATCH	/offers/:id/submit	H	Submit offer for approval (status = Pending Approval; creates ApprovalRequest)
PATCH	/offers/:id/send	H	Send approved offer PDF to candidate with expiring signing link (status = Sent)
PATCH	/offers/:id/expire	H, A	Manually mark an offer as Expired
DELETE	/offers/:id	H	Delete a Draft offer only (cannot delete offers in approval or sent state)
GET	/offers/:id/pdf	A, H	Generate and download offer PDF from template with all placeholders filled

2.5 Letter Generation

Method	Endpoint	Access	Description
GET	/letters	A, H	List all generated letters with type, version, and status
POST	/letters	H	Generate a letter by selecting employee + template; fills placeholders from profile
GET	/letters/:id	A, H, In, E	Get letter metadata and download URL
GET	/letters/:id/pdf	A, H, In, E	Download the generated letter PDF file
PATCH	/letters/:id/submit	H	Submit letter for approval (creates ApprovalRequest; status = Pending Approval)
PATCH	/letters/:id/regenerate	H	Regenerate letter PDF (increments version; old PDF retained in storage)
DELETE	/letters/:id	H, A	Delete a draft letter (cannot delete sent or signed letters)

2.6 Document Verification

Method	Endpoint	Access	Description
GET	/documents	A, H	List all uploaded documents with status filter
POST	/documents	H, C, In, E	Upload a document against a candidate or employee (PDF/JPG/PNG ≤5MB)
GET	/documents/:id	A, H, C*, In*, E*	Get document metadata and file download URL
GET	/documents/:id/file	A, H, C*, In*, E*	Download the actual document file from storage
PATCH	/documents/:id/verify	H	Mark document as Verified (status = Verified; notifies uploader)
PATCH	/documents/:id/reject	H	Reject document with a reason (status = Rejected; notifies uploader)
DELETE	/documents/:id	H, A	Delete a document record and file from cloud storage

2.7 Onboarding Tracker

Method	Endpoint	Access	Description
GET	/onboarding	A, H	List all onboarding checklists with progress percentage
GET	/onboarding/:employeeId	A, H, In, E	Get checklist with all tasks, owners, due dates, and statuses
POST	/onboarding	H, System	Create onboarding checklist for a new joiner (auto-triggered on offer acceptance)
PATCH	/onboarding/:id/tasks/:taskId	H, In, E	Update task completion status (triggers progress % recalculation)
POST	/onboarding/:id/tasks	H	Add a custom task to an existing checklist
DELETE	/onboarding/:id/tasks/:taskId	H	Remove a task from the checklist

2.8 Exit Management

Method	Endpoint	Access	Description
GET	/exits	A, H	List all active exit records with clearance progress and LWD
POST	/exits	H, E	Initiate a resignation (creates Exit record + ApprovalRequest for P1 flow)
GET	/exits/:id	A, H, E	Get full exit record with resignation date, LWD, reason, and clearance status
PATCH	/exits/:id/clearance/:dept	H	Mark a department clearance as complete (dept: it/finance/manager/hr)
PATCH	/exits/:id/approve	A, H	Approve a resignation request (P1 approval flow)
POST	/exits/:id/letter	H	Generate relieving + experience letters (only when all 4 clearances = complete)
PATCH	/exits/:id/complete	H	Mark exit as completed and set Employee status = Exited

2.9 HR Dashboard

Method	Endpoint	Access	Description
GET	/dashboard/hr	H, A	Return pipeline stats, pending approvals, open onboardings, and expiring offers
GET	/dashboard/hr/pipeline	H, A	Breakdown of candidates per pipeline stage
GET	/dashboard/hr/exports	H, A	Export HR summary data (candidates, employees, offers) as CSV or Excel

3. Hiring Manager APIs

Hiring Manager sees only assigned candidates and department hiring data. Role: HM

3.1 HM Dashboard

Method	Endpoint	Access	Description
GET	/dashboard/hiring-manager	HM	Return open roles, pipeline count, interviews this week, and offers to approve
GET	/dashboard/hiring-manager/roles	HM	List open roles assigned to this HM's department with candidate counts

3.2 Candidate Review

Method	Endpoint	Access	Description
GET	/hiring-manager/candidates	HM	List only candidates assigned to this HM's department (salary stripped)
GET	/hiring-manager/candidates/:id	HM	Get candidate profile + resume + interviewer feedback (no salary/offer data)
POST	/hiring-manager/candidates/:id/recommend	HM	Submit hiring recommendation (select/reject) with comments

3.3 Offer Approvals

Method	Endpoint	Access	Description
GET	/hiring-manager/approvals	HM	List all offer/letter approval requests pending this HM's action
GET	/hiring-manager/approvals/:id	HM	Get approval request details including offer and previous step decisions
PATCH	/hiring-manager/approvals/:id/approve	HM	Approve the current step (triggers next step or marks fully approved)
PATCH	/hiring-manager/approvals/:id/reject	HM	Reject with reason (returns request to HR for modification)

4. Interviewer APIs

Interviewer sees only assigned rounds. Salary/offer data is stripped at API level. Role: I

4.1 My Interviews

Method	Endpoint	Access	Description
GET	`/interviewer/interviews`	I	List all interview rounds assigned to this Interviewer (upcoming + past)
GET	`/interviewer/interviews/:roundId`	I	Get details of a specific round including candidate summary (no salary/offer)

4.2 Feedback Submission

Method	Endpoint	Access	Description
POST	`/candidates/:id/interviews/:roundId/feedback`	I	Submit rating (1–5 stars) and feedback text for an assigned round
PATCH	`/candidates/:id/interviews/:roundId/feedback`	I	Update feedback/rating before HR locks the round
GET	`/candidates/:id/interviews/:roundId/feedback`	I, H, HM	Get submitted feedback for a specific interview round

5. Candidate APIs

Candidate sees only own application, documents, and offer. Role: C

5.1 Application Status

Method	Endpoint	Access	Description
GET	/candidate/application	C	Get own application status with pipeline stage timeline and dates
GET	/candidate/profile	C	Get own candidate profile (name, position, applied date)

5.2 Document Upload

Method	Endpoint	Access	Description
GET	/candidate/documents	C	List own uploaded documents with verification status per document
POST	/candidate/documents	C	Upload a document (PDF/JPG/PNG ≤5MB); validates file type and size
PATCH	/candidate/documents/:id	C	Replace/re-upload a rejected document with a new file
DELETE	/candidate/documents/:id	C	Delete own pending document (cannot delete already-verified docs)

5.3 Offer & E-Signature

Method	Endpoint	Access	Description
GET	/candidate/offer	C	Get own offer details (visible only after offer status = Sent)
GET	/sign/:token	Public	Verify signing token validity and return offer PDF for review
POST	/sign/:token/accept	Public	Submit drawn signature + accept offer (records timestamp + IP; triggers role transition)
POST	/sign/:token/decline	Public	Decline the offer (sets offer status = Rejected)

6. Intern APIs

Intern sees own profile, onboarding checklist, documents, and letters only. Role: In

6.1 Onboarding Checklist

Method	Endpoint	Access	Description
GET	/intern/onboarding	In	Get own onboarding checklist with tasks, owners, due dates, and progress %
PATCH	/intern/onboarding/tasks/:taskId	In	Mark own assigned task as complete (recalculates progress)

6.2 Profile

Method	Endpoint	Access	Description
GET	/intern/profile	In	Get own intern profile (department, stipend, mentor, joining date, location)

6.3 Documents & Letters

Method	Endpoint	Access	Description
GET	/intern/documents	In	List own uploaded documents with current verification status
POST	/intern/documents	In	Upload a document for onboarding verification (PDF/JPG/PNG ≤5MB)
PATCH	/intern/documents/:id	In	Re-upload a rejected document with a corrected file
GET	/intern/letters	In	List own letters (Appointment, Internship Offer, etc.) — read-only
GET	/intern/letters/:id/pdf	In	Download own letter PDF file

7. Employee APIs

Employee sees own profile with tenure/salary, documents, letters, and can initiate resignation. Role: E

7.1 Employee Portal

Method	Endpoint	Access	Description
GET	/employee/profile	E	Get own full profile: tenure, CTC, manager, department, designation
GET	/employee/profile/history	E	Get own tenure history (promotions, transfers, increments with old/new salary)
GET	/employee/profile/salary	E	Get own salary breakdown (Basic, HRA, allowances, PF, total CTC)

7.2 Document Vault

Method	Endpoint	Access	Description
GET	/employee/documents	E	List own documents (onboarding uploads, verification docs, and company letters)
POST	/employee/documents	E	Upload a new document to own vault
PATCH	/employee/documents/:id	E	Re-upload a rejected document with a corrected file
GET	/employee/letters	E	List all own letters (offer, appointment, promotion, increment — versioned)
GET	/employee/letters/:id/pdf	E	Download a specific letter PDF by version

7.3 Resignation

Method	Endpoint	Access	Description
GET	/employee/resignation	E	Get own active resignation status and clearance progress
POST	/employee/resignation	E	Submit a resignation with reason and last working day request
DELETE	/employee/resignation	E	Withdraw resignation (only if not yet approved/in-clearance)

8. Shared APIs (All Authenticated Roles)

8.1 Notifications

Method	Endpoint	Access	Description
GET	/notifications	All	Get own notification list with read/unread status and total unread count
PATCH	/notifications/:id/read	All	Mark a single notification as read
PATCH	/notifications/read-all	All	Mark all own notifications as read
DELETE	/notifications/:id	All	Delete a single notification from own list

8.2 Approval Requests

Method	Endpoint	Access	Description
GET	/approvals	A, H, HM	List approval requests pending current user's action
GET	/approvals/:id	A, H, HM	Get full approval request with all steps, decisions, and comments
PATCH	/approvals/:id/approve	A, H, HM	Approve current step (per workflow config; triggers next step automatically)
PATCH	/approvals/:id/reject	A, H, HM	Reject with mandatory comment (returns to creator for modification)
GET	/approvals/history	A, H	List all completed approval requests (approved/rejected)

9. System / Internal Routes (node-cron)

These routes are triggered internally by middleware or node-cron. Not user-facing.

Method	Endpoint	Access	Description
POST	`/system/offers/:id/expire`	node-cron	Auto-expire offers past their expiry date; notify HR
POST	`/system/signatures/:token/expire`	node-cron	Auto-expire unsigned signing links past token expiry
POST	`/system/reminders/approvals`	node-cron	Send reminder notifications for approvals pending >24h
POST	`/system/reminders/onboarding`	node-cron	Send due-date reminders for pending onboarding checklist tasks
POST	`/system/reminders/exits`	node-cron	Send clearance reminders to departments that haven't cleared an exit
POST	`/system/role-transition`	Internal	Atomically convert Candidate → Intern/Employee on offer acceptance (DB transaction)

10. Access Control Matrix

☑ Full access · ◍ Scoped/partial · ✕ No access

Resource	Admin	HR	HM	Interviewer	Candidate	Intern	Employee
Users & Roles	☑ Full	✕	✕	✕	✕	✕	✕
Workflows Config	☑ Full	🟡 Read	✕	✕	✕	✕	✕
Letter Templates	☑ Full	🟡 Read	✕	✕	✕	✕	✕
Audit Log	☑ Read	🟡 Read	✕	✕	✕	✕	✕
All Candidates	☑ Full	☑ Full	🟡 Assigned	🟡 Assigned	🟡 Own	✕	✕
All Employees	☑ Full	☑ Full	✕	✕	✕	🟡 Own	🟡 Own
Offers	☑ Full	☑ Full	🟡 Approve	✕	🟡 Own+Sign	✕	✕
Generated Letters	☑ Full	☑ Full	✕	✕	✕	🟡 Own read	🟡 Own read
Documents	☑ Full	☑ Full	✕	✕	🟡 Own upload	🟡 Own upload	🟡 Own upload
Approval Requests	☑ Approve	🟡 Submit+Approve	🟡 Approve	✕	✕	✕	✕
Onboarding Checklist	☑ Full	☑ Full	✕	✕	✕	🟡 Own tasks	🟡 Own tasks
Exit / Resignation	☑ Full	☑ Full	✕	✕	✕	✕	🟡 Initiate own
Notifications	🟡 Own	🟡 Own	🟡 Own	🟡 Own	🟡 Own	🟡 Own	🟡 Own
Global Search	☑ All scope	🟡 Full scope	✕	✕	✕	✕	✕
Dashboard	☑ Admin view	🟡 HR view	🟡 HM view	✕	✕	✕	✕

11. Key Business Rules (API-Level Enforcement)

These rules must be enforced in backend middleware/service layer — not just the frontend.

Rule	Endpoint(s)	Enforcement
Offer send gate	`PATCH /offers/:id/send`	Block send if ApprovalRequest.status ≠ approved
Exit letter gate	`POST /exits/:id/letter`	Block generation unless all 4 clearances (IT/Finance/Manager/HR) = complete
Salary/PII stripping	`GET /candidates/:id (Interviewer)`	Strip salary, offer, offerDetails fields from response at middleware level
Token expiry check	`GET /sign/:token`	Validate token existence and expiry before serving offer PDF
Atomic role transition	`POST /sign/:token/accept`	Wrap 4 writes in MongoDB session/transaction (User + Employee + Checklist + AuditLog)
AuditLog on every mutation	`All POST / PATCH / DELETE`	Audit middleware writes to AuditLog with old/new values before returning response
Last Admin guard	`PATCH /users/:id/deactivate`	Reject if deactivating the only remaining active Admin account
Interviewer data scope	`GET /interviewer/interviews`	Return only rounds where interviewerId = current user._id
Candidate offer visibility	`GET /candidate/offer`	Return 404 if offer status is Draft, Pending Approval, or Rejected
Draft-only edit/delete	`PATCH/DELETE /offers/:id`	Reject modification if offer status ≠ Draft
Finance role clarity	`Workflows config`	Finance must map to a valid user role or department tag — not an undefined role
Resign approval gate	`POST /employee/resignation`	Create ApprovalRequest (P1); do not directly create Exit record on submission

12. Common Response Formats

12.1 Standard HTTP Status Codes

Code	Meaning
`200 OK`	Success (GET, PATCH)
`201 Created`	Resource created (POST)
`204 No Content`	Deleted successfully (DELETE)
`400 Bad Request`	Validation error or malformed request body
`401 Unauthorized`	Missing or invalid JWT token
`403 Forbidden`	Authenticated but insufficient role/permissions
`404 Not Found`	Resource does not exist or is not accessible
`409 Conflict`	Duplicate record (e.g., email already exists)
`422 Unprocessable Entity`	Business rule violation (e.g., offer not approved yet)
`500 Internal Server Error`	Unexpected server error

12.2 Response Body Shapes

Success (single): `{ "success": true, "data": { ... }, "message": "Operation successful" }`

Success (list): `{ "success": true, "data": [ ], "pagination": { "total": 48, "page": 1, "limit": 20, "totalPages": 3 } }`

Error: `{ "success": false, "error": { "code": "FORBIDDEN", "message": "You do not have permission to access this resource" } }`